

SDV Fault-Prediction & Self-Healing

Simulation-Only Sprint Roadmap · mid-August → 30 September 2026

Premise. No physical board is on any critical path (ADR-012). The full July–December plan plus the Jan–Jul 2027 margin is compressed into a ~6.5-week full-time sprint that delivers **all eight architecture layers (L0–L7) end-to-end in simulation** — Renode for the vehicle, a local GPU for the LLM, and Docker/Railway for the cloud plane. The hard architecture rule is preserved throughout: **on-vehicle AI may act · cloud AI only proposes · off-vehicle AI only explains**; the plain-C supervisor holds the veto. New elements requested — the **LLM reporter on the local GPU** (L6 / agent E) and **n8n automations** (off-vehicle orchestration) — both sit off-vehicle and never actuate.

Honesty note. This timeline is aggressive. The vertical slice (on-MCU int8 inference) is front-loaded to Sprint 2 so the December-style go/no-go surfaces in August, not September. The novel items (cross-ECU GNN beating an MLP; LLM-RAG report quality) carry the real risk and each has a named fallback.

Architecture layers being delivered (all in simulation)

Layer	Plane	What it is	Status now
L0 substrate	On-vehicle	S32K388 platform + UART-hub bus (CAN stand-in, ADR-017)	DONE
L1 sense + inject	On-vehicle	Heap/timing/acoustic signals; fault-control block hooks	leak done; timing+acoustic next
L2 on-vehicle AI	On-vehicle	int8 autoencoder analyst + TTF predictor on the M7	to build
L3 supervisor + ACT	On-vehicle	Plain-C veto + whitelist heal (RESTART/DEGRADED/LOAD_SHED)	to build
L4 data backbone	Off-vehicle	MQTT → Redpanda → TimescaleDB + pgvector → FastAPI	to build
L5 cloud advisory	Off-vehicle	Cross-ECU GNN v0 (proposes which ECU degrades next)	to build
L6 reporting + HITL	Off-vehicle	Grafana + React panel + LLM reporter (local GPU) + n8n	to build
L7 research brain	Meta	Obsidian vault (source of truth); this roadmap	DONE / live

Two spines run through every layer: **Safety & Certification** (ISO 26262 / SOTIF — framing, not certification) and **DevOps/MLOps** (Git · Docker · CI · DVC · W&B · pytest).

The seven sprints

Sprint 0 — Data foundation complete		Aug 13–17	L1 - MLOps
Work	Multi-run memory-leak capture (rates 64/128/256, ≥ 5 seeds). Wire the timing / deadline-miss injection (busy_spin_us, already in the fault-control block) → labelled dataset. Ingest the MIMII acoustic dataset → feature CSV (replay path). DVC init + dataset card.		
Exit	Three fault classes have labelled, DVC-tracked datasets; dataset card written.		
Sprint 1 — Baseline ML		Aug 18–24	L2
Work	ml/dataset.py (pivot long→wide, rolling-slope features, time-series splits only). scikit-learn Isolation Forest + PyTorch denoising autoencoder baselines. Predictor: heap-slope → TTF regression scored against the analytical true-TTF. W&B run logging.		
Exit	Baseline ROC-AUC + false-positive/hour per class ; predictor lead-time MAE reported.		
Sprint 2 — On-target int8 (THE VERTICAL SLICE)		Aug 25–31	L2
Work	int8 post-training quantization + calibration set; litert-torch export with a model.eval() parity check. Run the int8 autoencoder on the emulated Cortex-M7 in Renode (LiteRT-Micro / CMSIS-NN). Emit the anomaly score over UART; measure tensor-arena KB and indicative latency.		
Exit	Injected fault → int8 inference on S32K3 in Renode → anomaly score. GO/NO-GO checkpoint .		
Sprint 3 — Close the loop		Sep 1–7	L3
Work	Plain-C safety supervisor: whitelist actions, veto logic that preempts inference. Wire it into K3 handle_link_line(): detect → predict → supervisor-approved RESTART before OOM .		
Exit	Memory-leak class healed end-to-end in sim; supervisor veto count > 0 (the cage is live).		
Sprint 4 — Data backbone + dashboard		Sep 8–14	L4 - L6a
Work	MQTT (Mosquitto) → Redpanda → TimescaleDB + pgvector → FastAPI, in Docker (Railway optional). K3 uplink bridge → MQTT. Grafana live dashboard. n8n workflows for incident routing + alerting.		
Exit	Injected fault visible on Grafana within seconds; incident stored; n8n alert fires.		
Sprint 5 — GNN + LLM reporter		Sep 15–21	L5 - L6b
Work	Cross-ECU GNN v0 (2-layer GCN, 5-node graph), advisory — predict which ECU degrades next, must beat an MLP baseline (top-1/top-2 node accuracy). LLM reporter on the local GPU (Qwen + RAG over pgvector) — explains incidents, never actuates. n8n triggers the reporter on each new incident.		
Exit	GNN emits a propagation proposal; LLM produces a retrieved incident report.		
Sprint 6 — Measure + assemble		Sep 22–30	Safety - Metrics - Demo
Work	Metrics campaign over ≥ 30 injected incidents: MTTR, recovery success rate, FP/hour, ROC-AUC, the float32-vs-int8 size/latency/AUC table, veto count. ISO 26262 / SOTIF framing write-up. Minimal React HITL panel. One-command demo; thesis chapters drafted from the vault.		
Exit	End-to-end demo runnable from one command; metrics table complete; chapters in draft.		

Definition of Done — 30 September 2026

A single command boots the simulation and runs the complete loop: **inject fault** (L1) → on-MCU **int8 detect** (L2) → **predict TTF** (L2) → **supervisor heals** before failure (L3) → **uplink** (L4) → **Grafana + n8n alert** (L6) → **GNN proposal** (L5) → **LLM report** from your GPU (L6). The metrics table is complete and every number traces to a dated run ID + injection seed.

Risks & fallbacks

Risk	Fallback (decided now, not in the last week)
Timeline is full-time and dense	Sprints 0–3 are the protected core (data → baseline → on-MCU → heal). If time slips, L5/L6 shrink to demos; the closed loop ships.
GNN must beat the MLP baseline	If it doesn't, the GNN ships as an advisory demo and the MLP is the reported cross-ECU result — no overclaim.
LLM-RAG report quality on local GPU	Fallback to a templated incident report populated from pgvector; the LLM becomes the polish layer, not a dependency.
On-target latency not cycle-accurate	Report ms as indicative; hard numbers wait for the MR-CANHUBK344 board (out of this scope).
CAN bus-off / bus realism	Not simulable (decorative registers). Graduated to the board; timing/deadline-miss is the second sim flagship.

Generated for the SDV Fault-Prediction & Self-Healing thesis (Maior Cristian-Alexandru, ULBS). Authoritative decisions live in the Obsidian vault (30-Architecture/ADRs). This file is mandated reading by CLAUDE.md RULE 2.